Flight Software Workshop Agenda
Day 1 - Monday, February 9th 2026, 8:30 am - 4:00 pm

Type	Topic/Activity	Presenter	Time	Duration
Introduction	00 Introduction	Jeff Levison	8:30	15
Presentation	01 Spacecraft Archtecture and System Overview	Tim Canham	8:45	15
Presentation	02 FSW System Engineering - Capability Layers Services Applications Drivers	Tim Canham	9:00	30
Presentation	03 FSW Architecture, Requirements & Design Software architecture Software requirements Structural analysis Data flow diagrams Software design State machine models	Garth Watney	9:30	60
Break			10:30	15
Presentation	04 F' Introduction	Tim Canham	10:45	75
Lunch			12:00	60
	05 FPP Introduction	Rob Bocchino	1:00	15
Exercise	Lab project introduction	Michael Starch	1:15	15
	Lab project setup	Michael Starch	1:30	15
	System requirements definition	Michael Starch	1:45	30
Break			2:15	15
Presentation	06 FSW Design FSW system modeling in F' (Types, ports, components, topology definitions) Data types, interfaces Commands, telemetry events, parameters and data products Concurrency, - initialization, memory allocation OS abstraction layer (message queues, mutex, tasks, file system) Interface design patterns Serializing and deserializing data Designing and visualizing topologies	Michael Starch	2:30	90

Day 2 - Tuesday, February 10th, 2026, 8:30 am - 3:30 pm				
Type	**Topic/Activity**	**Presenter**	**Time**	**Duration**
Exercise	Component Design and Initial Implementation Implementing commands and events	T. Boyer-Chammard	8:30	60
Exercise	Initial Component Integration F´ Deployment and introduction to the GDS Establishing the initial topology	T. Boyer-Chammard	9:30	30
Break			10:00	15
Presentation	07 Reducing Risk Defensive coding Avoiding C and C++ pitfalls Static and dynamic analysis	Rob Bocchino	10:15	45
Presentation	08 Suggestions for Coding Style Modules and components Functions Expressions and statements	Rob Bocchino	11:00	45
Presentation	09 Data Structures Library Overview Review of basic data structures for FSW Overview of F Prime data structure library	Rob Bocchino	11:45	15
Lunch			12:00	60
Exercise	Full Component Implementation Implementing telemetry, parameters, port calls	Kevin Ortega	1:00	60
Exercise	System Integration Completing the topology	Kevin Ortega	2:00	30
Exercise	Running on Hardware	Kevin Ortega	2:30	60
Lab Tour (optional)			3:30	60

Day 3 - Wednesday, February 11th, 2026, 8:30 am - 4:00 pm				
Type	**Topic/Activity**	**Presenter**	**Time**	**Duration**
Presentation	10 Unit Testing Basic principles of testing Writing modular tests Achieving and checking code coverage Picking good inputs	Rob Bocchino	8:30	30
Presentation	11 F´ Unit Test Framework	Rob Bocchino	9:00	30
Exercise	Writing unit tests	Rob Bocchino	9:30	60
Break			10:30	15
Presentation	12 Systems Testing F' GDS overview Test API	Michael Starch	10:45	45
Lunch			11:30	60
Exercise	System testing with the GDS	Celeste Smith	12:30	65
Presentation	13 FSW Development Process FSW level process - waterfall model, artifacts, reviews Component level process - checklists, reviews FSW planning and reporting	Aadil Rizvi	1:35	60
Presentations	Lightning Talks	Various	2:35	85
Closing		Jeff Levison	4:00	5

Lightning Talks - Agenda			
Affiliation	**Title**	**Presenter**	**Time**
Johns Hopkins University	OrbitDSP: Integrated Signal Processing and Orbital Dynamics for Embedded Systems	Herbert Beaumont	2:40
JPL	Planetary Landing Autopilot	Andrew Johnson	2:45
Texas State University	PROVES kit: The Open-Source CubeSat	Saidi Adams Aaron Siemsen	2:50
Blue Eye Soft	Flight Software Challenges for Space Weather–Aware Autonomous Systems	Srikanth Kodeboyina	2:55
JPL	Hermes: A Command & Telemetry Framework	Andrei Tumbar	3:00
Aclectic Systems Inc.	SynthetixTM and PolymathTM, A Work-In-Progress Towards A Digital Twin Authoring Environment and Edge Computing Platform for Spacecraft Systems	Yahya H. Mirza	3:05
Varda Space Industries	Design and Evaluation of a Lightweight On-Orbit Probe for the NASA GPDM Satellite	Atilla Saadat	3:10
University of Washington	HuskySat-2: Autonomous Optical Navigation for CubeSats	Arin Kharkar Yaj Jhajhria David Magidson	3:15
JPL	HPSC and ENGAGE Avionics Platforms	Eli McMahon	3:20
Utah State University	Leveraging F' V4 Subtopologies for a Custom Communication Protocol	Boston Larson Gavin Steel Ethan Wayland	3:25
Yonsei University	CubeSat activities at Yonsei University	Youngho Eun	3:30
JPL	Rover Operations Center	Jennifer Trosper	3:35
Cal Poly Pomona	CADENCE: Designing Flight Software for the University Nanosat Program Using F'	Michael Pham	3:40
Oligo Space	Oligo Space - F' in Context	Samuel Villarreal	3:45
JPL	New JPL Option for Personalized Industry and Academia Support	Benji Donitz	3:50
Flapmax	Bounded Reconfiguration Under Flight Software Authority for Autonomous Small Spacecraft	Dave Ojika	3:55

Ordering subject to change